

AI Hub & Signal Processing Club

Mbeya University of Science and Technology

WEBSITE DEVELOPMENT BRIEF & TECHNICAL SPECIFICATION

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Prepared for prospective developers / development agency

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Table of Contents

Table of Contents.....	2
1. Project Overview	4
1.1 Goals of the New Website	4
1.2 Out of Scope (for this version)	4
2. Target Audience.....	4
3. Information Architecture & Sitemap.....	5
3.1 Secondary / Utility Pages.....	5
4. Page-by-Page Feature Requirements	6
4.1 Home Page	6
4.2 About Page.....	6
4.3 Sub-Groups Pages	6
4.4 Research Page.....	6
4.5 Projects Page	7
4.6 Events Page	7
4.7 Blog / News	7
4.8 Resources	7
4.9 Members Directory	7
4.10 Join Us	8
4.11 Contact.....	8
5. AI Assistant (Chatbot)	8
5.1 Purpose & Use Cases	8
5.2 Recommended Approach	8
5.3 Acceptance Criteria for the Chatbot.....	9
6. Content Management & Admin Requirements	9
6.1 Required CMS Capabilities.....	9
6.2 CMS Approach Recommendation.....	9
7. Recommended Technical Stack.....	10
7.1 Non-Functional Requirements	11
8. Design & Branding Guidelines	11
8.1 Visual Direction: "AI for Good"	11
8.2 Brand Consistency	12
9. Integrations	12
10. Content & Assets the Club Will Supply.....	13
11. Suggested Timeline & Milestones	13
12. Deliverables & Acceptance Criteria	13

12.1 Final Deliverables	13
12.2 Acceptance Criteria	14
13. Budget & Engagement Notes	14
14. Project Contact	14

1. Project Overview

The AI Hub & Signal Processing Club is a student-led technical organization at Mbeya University of Science and Technology (MUST), based in the Department of Electronics and Telecommunication Engineering / Faculty of Computing and Information Technology. The club has 150+ student members organized into 7 technical sub-groups and is coordinated by Jofrey Nasson Msengi.

This document is the complete brief and specification for a new version of the club's official website. It is intended to be handed directly to a developer or development agency to scope, quote, and build the project without requiring further clarification on core requirements.

1.1 Goals of the New Website

- Establish a credible, professional public face for the club that reflects its standing in AI, signal processing, and broader STEM at MUST and within Tanzania's growing tech ecosystem.
- Showcase the club's 7 technical sub-groups, ongoing projects, research output, and achievements (e.g. hackathons, IEEE contest entries, conference selections).
- Make it easy for current MUST students to discover the club, learn what each sub-group does, and apply to join online.
- Give the club a channel to publish news, events, blog posts/tutorials, and recaps that can also be cross-posted to LinkedIn.
- Provide a lightweight, secure admin area so non-technical student leaders can update content (events, news, members, projects) without developer involvement after launch.
- Support partnership and funding conversations by presenting the club as an organized, well-documented institution (useful alongside grant work such as the NIHR AMR surveillance application and other proposals).

1.2 Out of Scope (for this version)

- E-commerce / merchandise sales.
- A dedicated mobile app (the site must be fully responsive instead).

2. Target Audience

Audience	What they need from the site
Prospective MUST students	Quick understanding of what the club does, which sub-group fits their interest, and an online application/join form.
Current members	Event schedules, project pages, resources/learning materials, and recognition (member/contributor profiles).
MUST faculty & administration	Evidence of organized activity, achievements, and impact to support institutional support and approvals.
External partners, funders, judges	Credibility signals: leadership bios, past projects, publications, media mentions, and a way to contact the club.

Audience	What they need from the site
General public / press	News, blog posts, and a clear "About" story of the club's mission and impact in Mbeya/Tanzania.

3. Information Architecture & Sitemap

Primary navigation (top menu):

1. Home
2. About — Mission, history, leadership, faculty advisor(s), affiliation with MUST
3. Sub-Groups — overview grid of the 7 technical sub-groups, each with its own sub-page
4. Research — the club's Research Team, active research lines, publications, and collaborators
5. Projects — portfolio of past and current projects (filterable by sub-group/technology)
6. Events — upcoming and past events, hackathons, workshops, with calendar view
7. Blog / News — articles, tutorials, recaps, written by members and coordinators
8. Resources — learning materials, slide decks, datasets, links, recommended courses
9. Members — public member directory / contributor wall (opt-in, photo + role + sub-group)
10. Join Us — online application form + FAQ + membership requirements
11. Contact — contact form, email, social links, location/map

3.1 Secondary / Utility Pages

- Sub-group detail pages (7 total) — e.g. Machine Learning, Signal Processing, Web developers .
- Individual research line / publication detail page (template, reused per research output).
- Individual project detail page (template, reused per project).
- Individual blog post page (template).
- Individual event detail page (template, with RSVP if applicable).
- Site-wide AI Assistant (chatbot) widget — appears as a floating launcher on every page, not a standalone page.
- 404 / Not Found page.
- Privacy Policy & Terms (basic, since the join form and contact form collect personal data).
- Admin login (not in public nav).
- Real-time chat/community platform (Slack/Discord/WhatsApp links will be used instead and simply linked from the site).
- Multi-language UI beyond English (Swahili content can be added as blog posts, but full UI localization is not required in v1).

4. Page-by-Page Feature Requirements

4.1 Home Page

- Hero section: club name, tagline reflecting an "AI for Good" / impact-driven mission, hero image or short video/animation, primary call-to-action buttons ("Join the Club", "View Projects").
- Hero should subtly surface the AI Assistant (chatbot) launcher, e.g. a prompt like "Ask our AI Assistant about the club" to reinforce that this is an AI-first organization, not just a generic club site.
- Quick stats strip: number of members, number of sub-groups, number of completed projects, number of research publications, founding year.
- Sub-groups preview grid (icon + short description, links to sub-group pages).
- Featured/recent projects carousel or grid (3–6 items).
- Upcoming events list (next 3 events, pulled dynamically from Events module).
- Latest blog posts (3 most recent, pulled dynamically).
- Partner/affiliate logos strip (MUST logo, any sponsors/partners).
- Newsletter / mailing list signup (email capture).
- Footer: quick links, social icons (LinkedIn, X, Instagram, GitHub), contact email, copyright.

4.2 About Page

- Mission & vision statement.
- Club history/timeline (founding, milestones).
- Leadership team section: Coordinator, sub-group leads, Research Team lead/Principal Investigator, faculty advisor — photo, name, role, short bio, optional LinkedIn link.
- Affiliation statement (MUST, Department of Electronics & Telecommunication Engineering / Faculty of Computing and Information Technology).

4.3 Sub-Groups Pages

- Overview page: grid/card layout of all 7 sub-groups with icon, name, 1–2 line description.
- Each sub-group detail page: full description, focus areas, lead(s), member count, related projects (auto-pulled by tag), related blog posts, and a "join this group" CTA.

4.4 Research Page

- Dedicated "Research Team" page distinct from general Sub-Groups and Projects — positions the club as a serious academic/research body, not just a hobbyist club.
- Overview of the Research Team: mission, current research lines (e.g. AI for health surveillance, AMR surveillance, NLP for low-resource African languages such as Bena and Nyakyusa, signal processing applications), and lead researchers/PI (e.g. Dr. Charles Okanda Nyatega as principal investigator collaborator and Joseph Laurent).

- Publications & outputs list: papers, preprints, technical reports, conference/contest entries — each with title, authors, venue/date, abstract, and link (DOI, PDF, or GitHub repo).
- Active grants/collaborations section (can be kept high-level/public-safe): partner institutions, funding bodies where appropriate to disclose publicly, and research focus areas.
- Call-to-collaborate section: how external researchers, institutions, or students can propose collaboration or join the Research Team.
- Research detail page template: full abstract, methodology summary, team/authors, related projects, downloadable PDF where available, and citation block.

4.5 Projects Page

- Grid/list of projects with thumbnail, title, sub-group tag, technology tags, status (ongoing/completed).
- Filter/search by sub-group and technology tag.
- Project detail page: description, problem statement, technologies used, team members, images/screenshots, links (GitHub repo, demo, paper/report), outcomes/results.
- Should support showcasing items such as the IEEE Myron Zucker water-quality monitoring entry, GSoC/DIPY contributions, and hackathon projects as examples of content types.

4.6 Events Page

- Upcoming events list with date, time, location (physical or online), short description, and registration link or button (can link to a Google Form initially).
- Past events archive with recap text, photo gallery, and outcomes.
- Optional: calendar view (month grid) in addition to list view.
- Event detail page template: full description, agenda, speakers, photo gallery, downloadable resources.

4.7 Blog / News

- List view with featured image, title, excerpt, author, date, category/tag.
- Categories: News, Tutorials, Research, Event Recaps, Member Spotlights.
- Individual post page: rich text content, images, embedded code blocks/snippets (for technical tutorials), author bio box, related posts, share buttons (incl. LinkedIn, given the club's active LinkedIn presence).
- Comment section optional (can be a simple moderated comment box or disabled in v1).

4.8 Resources

- Categorized list of resources: course links, datasets, tools, recommended reading, slide decks/PDFs for download.
- Should support file uploads (PDF/PPTX) and external links.

4.9 Members Directory

- Opt-in public profile grid: photo, name, sub-group, role, optional links (GitHub, LinkedIn).
- Filter by sub-group.
- Note: must be opt-in and GDPR-style consent captured at signup, since this is personal data of students.

4.10 Join Us

- Online application form: name, registration/student number, programme of study, year of study, email, phone, preferred sub-group, motivation statement, how they heard about the club.
- Form submissions stored in admin dashboard and emailed to the coordinator automatically.
- FAQ section (eligibility, meeting schedule, costs if any, expectations).

4.11 Contact

- Contact form (name, email, subject, message) with spam protection (e.g. reCAPTCHA/honeypot).
- Direct email and social links.
- Embedded map showing MUST campus, Mbeya.

5. AI Assistant (Chatbot)

Given the club's identity as an AI organization, the site should include a site-wide AI Assistant chatbot — this is both a functional support tool and a statement piece that demonstrates the club's own capability.

5.1 Purpose & Use Cases

- Answer visitor questions about the club: what it does, the 7 sub-groups, how to join, upcoming events, membership requirements.
- Help prospective members find the right sub-group based on their interests (a short guided Q&A).
- Surface relevant content: point users to specific projects, blog posts, research publications, or resources based on their question.
- Assist with FAQs for Join Us and Events (reducing repetitive email/DM questions to the coordinator).
- Optional stretch feature: lightweight assistant for members inside a logged-in area to query the Resources library ("what materials do we have on signal processing basics?").

5.2 Recommended Approach

- Use a retrieval-augmented generation (RAG) pattern: the chatbot is grounded in the site's own content (About, Sub-Groups, Research, Projects, Events, Resources, FAQ) rather than answering freely, to avoid hallucinated or off-brand answers.
- Suggested stack: a FastAPI endpoint that wraps the Anthropic Claude API or OpenAI API for the language model, with a vector store for retrieval (e.g. PostgreSQL with the

pgvector extension — keeping everything in the same Postgres database as the rest of the app — or a managed option like Pinecone), indexing the site's CMS content and refreshed whenever content is published/updated.

- Chat UI: a floating launcher bottom-right on all pages, expanding into a chat panel; must be fully usable on mobile.
- Always show a clear disclaimer that responses are AI-generated, plus an easy escalation path ("Still need help? Contact us") linking to the Contact page/email.
- Conversation logs should be stored (anonymized where possible) so club leadership can review common questions and improve content/FAQs over time.
- Cost control: set sensible rate limits per visitor/session and a monthly budget cap on API usage, since this is a student-run, budget-constrained project.

5.3 Acceptance Criteria for the Chatbot

- Chatbot correctly answers at least the core FAQ set (joining process, sub-groups, events, contact) without fabricating information not present on the site.
- Chatbot is available on every page, loads asynchronously, and never blocks core page rendering/performance.
- Admin can review chat logs and update the underlying content/FAQ without developer involvement.

6. Content Management & Admin Requirements

Student leadership changes year to year, so the site must be maintainable by non-developers after handover.

6.1 Required CMS Capabilities

- Role-based admin accounts: Super Admin (coordinator), Editor (sub-group leads), Contributor (members who can draft blog posts for approval).
- CRUD (create/read/update/delete) management for: Blog posts, Events, Projects, Research publications/lines, Sub-groups, Members directory entries, Resources, Home page featured content, and Join Us / Contact form submissions.
- Rich text editor with image upload, embedded links, and code-block support for technical posts.
- Media library for images/files with reasonable storage limits.
- Draft/Publish/Schedule workflow for blog posts and events.
- Simple analytics view or integration (page views, top content) — can be via Google Analytics embed rather than custom-built.

6.2 CMS Approach Recommendation

The club has decided on a FastAPI (Python) backend rather than a CMS like WordPress. This means a fully custom-built admin panel — full control over data models (Research, Projects, Sub-groups, Chatbot content, etc.) and a clean foundation for the AI Assistant/RAG layer, since

it lives in the same Python ecosystem, but it requires the development team to build the admin UI (rather than getting one for free from a CMS) and to commit to longer-term maintenance.

- Admin panel should be a separate authenticated React/Next.js app (or an admin section of the main frontend) talking to the FastAPI backend via REST or GraphQL — not server-rendered templates.
- Consider SQLAlchemy or FastAPI-Admin as a starting point for the admin UI to reduce build time, then customize as needed, rather than building every CRUD screen from scratch.
- Role-based access (Super Admin / Editor / Contributor) should be implemented via FastAPI's dependency injection + JWT/OAuth2 auth, with permissions enforced at the API level, not just hidden in the UI.

7. Recommended Technical Stack

Layer	Recommended option(s)	Notes
Frontend	Next.js (React) + TypeScript	Consumes the FastAPI backend via REST; enables SSR/SSG for good SEO and fast loads.
Styling	Tailwind CSS	Fast, consistent, easy to theme to club branding.
Backend / API	FastAPI (Python)	Powers all content APIs (Blog, Events, Projects, Research, Sub-groups, Members, Resources), the Join Us/Contact forms, auth, and the chatbot endpoint.
Admin panel	Custom React/Next.js admin (optionally bootstrapped with SQLAlchemy/FastAPI-Admin) talking to the FastAPI backend	Role-based access for Super Admin / Editor / Contributor.
ORM / Database	SQLAlchemy + Alembic migrations, PostgreSQL	PostgreSQL recommended over MySQL for better JSON/vector support, which also benefits the chatbot's RAG store.
Auth	FastAPI + OAuth2/JWT (e.g. fastapi-users library)	Covers both admin login and any future member login area.
Background jobs	Celery or FastAPI BackgroundTasks/ARQ with Redis	For sending notification emails, scheduled posts, and refreshing the chatbot's content index.
Hosting (backend)	VPS or PaaS supporting Python/Docker (e.g. Render, Railway, DigitalOcean, or a campus-hosted VPS)	Containerize the FastAPI app with Docker for portability and easy redeployment.
Hosting (frontend)	Vercel or Netlify for the Next.js app	Fast global CDN delivery; integrates cleanly with a separately hosted FastAPI backend.

Layer	Recommended option(s)	Notes
Domain & Email	A .ac.tz or .org/.com domain (e.g. mustaihub.org); Google Workspace or Zoho for club email	Confirm whether MUST IT department can issue a subdomain (e.g. aihub.must.ac.tz).
Forms	FastAPI endpoints + email delivery (e.g. via SMTP/SendGrid)	Join Us, Contact, and Newsletter forms post directly to the FastAPI backend, no separate form service needed.
Version control	GitHub (public repo for transparency, ties to club's open-source ethos)	Aligns with members' existing GitHub/open-source involvement; monorepo or separate frontend/backend repos both acceptable.
API documentation	FastAPI's built-in OpenAPI/Swagger docs	Should be kept available (at least to admins) as living technical documentation for future maintainers.

7.1 Non-Functional Requirements

- Performance: pages should load in under 3 seconds on a typical 3G/4G connection in Tanzania; images optimized/lazy-loaded; FastAPI endpoints should respond in well under 500ms for standard content reads.
- Mobile-first responsive design — majority of student traffic will be mobile.
- Accessibility: reasonable WCAG 2.1 AA effort (alt text, color contrast, keyboard navigation).
- SEO: clean URLs, meta tags, sitemap.xml, robots.txt, Open Graph tags for social sharing (especially LinkedIn, given the club's active page there); use Next.js SSR/SSG rather than pure client-side rendering so content is crawlable.
- Security: HTTPS by default, sanitized/validated form inputs (FastAPI + Pydantic validation), spam protection on public forms, regular database backups, admin accounts protected by strong passwords / optional 2FA, rate limiting on public API endpoints.
- Low-bandwidth resilience: avoid unnecessarily heavy assets; provide a lightweight experience for users on limited data plans.
- Browser support: latest 2 versions of Chrome, Safari, Firefox, Edge, plus mobile Chrome/Safari.

8. Design & Branding Guidelines

The site should feel premium, modern, and credible — closer to a serious AI research lab or a well-funded tech nonprofit than a typical campus club page. It should comfortably sit alongside the kind of institutions the club already engages with through grant and partnership work (e.g. NIHR, NumFOCUS, IEEE).

8.1 Visual Direction: "AI for Good"

- Lead with an "AI for Good" / impact narrative throughout the design: AI applied to real problems in Tanzania and East Africa — health (AMR surveillance), water quality, low-resource language NLP, education — rather than abstract or purely technical framing.
- Use real outcomes and people wherever possible (member and community photos, real project screenshots) instead of generic stock AI imagery (robots, glowing brains, matrix code) which reads as dated and impersonal.
- Subtle, tasteful tech motifs are welcome (signal waveforms, neural-network node patterns, circuit-trace lines) used as background texture/accents — not as the dominant visual language.
- Dark-mode-friendly, high-contrast "deep tech" palette (e.g. deep navy/charcoal base with a vivid accent color — electric blue, teal, or amber) reads as more modern and professional than light, generic corporate templates; final palette should still incorporate MUST's institutional colors where appropriate.
- Generous white space, large confident typography, and smooth (but tasteful) micro-interactions/animations on scroll — these are now baseline expectations for a "modern" site, not optional extras.

8.2 Brand Consistency

- Color palette, logo, and typography should be consistent with the club's existing LinkedIn banner/branding (to be supplied by the club) — developer should request brand assets before design begins.
- Provide a simple style guide deliverable (colors, fonts, logo usage, button styles, iconography) as part of the build, so future content stays consistent.
- Photography: prioritize real photos of members, events, research, and projects over generic stock imagery wherever possible.
- The Research and Projects sections in particular should be designed with extra visual weight/credibility (clean data visualizations, publication-style layouts) since these are the strongest "proof points" of the club's seriousness to funders and partners.

9. Integrations

Integration	Purpose
LinkedIn (company page)	Cross-link the site and LinkedIn page; "share to LinkedIn" buttons on blog/event posts.
Google Analytics / Plausible	Traffic and engagement tracking.
Google Forms (optional fallback)	Join Us / event RSVP if native forms are deprioritized in v1.
Email (Google Workspace/Zoho/SMTP)	Form notification delivery, mailing list.
GitHub	Link to club/members' open-source repos and project code.
AI Chatbot Engine	RAG-grounded answers about the club, sub-groups, research, events, and FAQs (see Section 5).
Google Maps	Embedded campus location on Contact page.

Integration	Purpose
reCAPTCHA / hCaptcha	Spam protection on public-facing forms.

10. Content & Assets the Club Will Supply

To avoid delays, the following must be provided by the club before/during development:

- Final club logo (vector/SVG preferred) and any existing brand guidelines.
- Final names and descriptions for the 7 technical sub-groups.
- Leadership bios and headshots (Coordinator, sub-group leads, Research Team lead/PI, faculty advisor).
- List and writeups of past/current projects (with images) for the Projects section.
- Research Team content: active research lines, publications/reports list (with PDFs/links), collaborator/partner institutions, and PI bio.
- Initial blog content (3–5 articles) to populate the site at launch.
- Event history (past events with photos) and a calendar of upcoming events.
- Club mission/vision statement and history/timeline text.
- Social media links and official contact email.

11. Suggested Timeline & Milestones

Phase	Duration	Deliverable
1. Discovery & content collection	1–2 weeks	Confirmed sitemap, brand assets, and content gathered from the club.
2. Design (wireframes + UI design)	1–2 weeks	Wireframes and high-fidelity mockups for key pages, approved by club leadership.
3. Development (frontend + CMS setup)	3–5 weeks	Functional site on staging environment with CMS populated by club content.
4. Testing & QA	1 week	Cross-browser/device testing, form testing, performance/accessibility checks.
5. Launch	2–3 days	Site live on production domain with SSL, analytics, and backups configured.
6. Handover & training	2–3 days	Admin training session/video + documentation for the club's student leaders.

12. Deliverables & Acceptance Criteria

12.1 Final Deliverables

- Fully responsive, deployed website on the agreed production domain with HTTPS.
- Admin/CMS access with all required content types populated at launch.
- Source code repository (GitHub) with README and setup instructions.
- Style guide / brand asset reference.
- Admin handover documentation (how to add a blog post, event, project, member, etc.) — ideally a short screen-recorded video plus a written guide.
- List of all third-party accounts/credentials created during the build (hosting, domain registrar, analytics, email service), transferred to club ownership.

12.2 Acceptance Criteria

- All pages in Section 3 (Sitemap) exist and are reachable from primary navigation.
- All forms (Join Us, Contact, Newsletter) successfully submit and notify the coordinator by email.
- Site scores reasonably on Google PageSpeed/Lighthouse (target: 80+ on mobile performance, 90+ on accessibility/SEO/best practices, acknowledging some flexibility for image-heavy pages).
- Admin can independently create/edit/delete a blog post, event, project, and research entry without developer assistance, verified in a live walkthrough before final sign-off.
- AI Assistant chatbot is live site-wide, answers the core FAQ set correctly, and is reviewable/maintainable by admin per Section 5.3.
- Site is fully usable on a mid-range Android phone over a 4G connection.

13. Budget & Engagement Notes

As a student club, budget is constrained. Recommended approach for the proposal request to developers:

- Since the backend stack (FastAPI) is now fixed, ask developers to quote against the stack defined in Section 7 rather than proposing alternative platforms — this keeps quotes comparable across candidates.
- Request a fixed-price quote broken down by phase (Section 10) rather than a single lump sum, to make trade-offs easier to negotiate.
- Clarify recurring costs separately from one-time build costs: domain registration, hosting, and any paid plugins/services should be itemized as annual recurring costs.
- Where possible, prioritize free/low-cost tiers for analytics, forms, and email at this stage, with a clear upgrade path noted for when membership and traffic grow.

14. Project Contact

All questions on this brief, content clarifications, and brand assets should be directed to:

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